

Performance Management System

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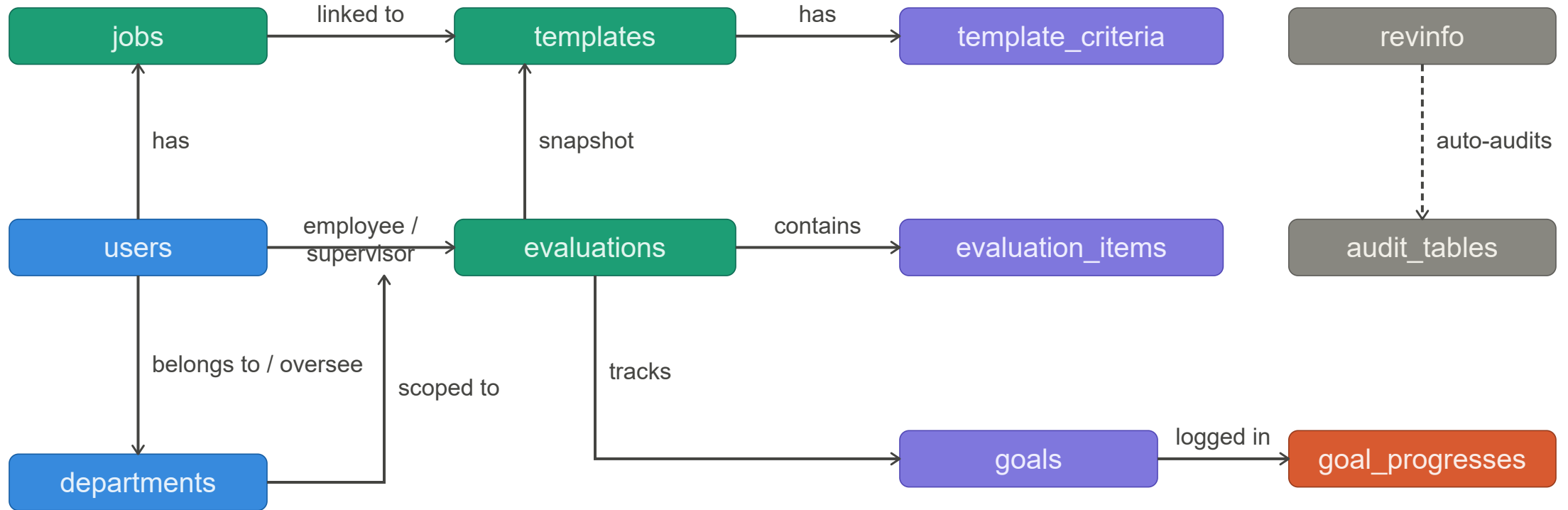
AC — Goal Management

Requirement	When	Then
SMART goal setting	Employee drafts a goal	All SMART fields enforced
Progress tracking	Employee add a progress	Linked to goal
Customizable templates	Manager opens a review	Shows HR-defined job criteria

AC — Performance Review & Compliance

Requirement	When	Then
Result dispute	Employee receive review	Must confirm or reject to proceed
RBAC	Access restricted resource	System denies
Immutable audit log	Save change	New entry appended

ER Model

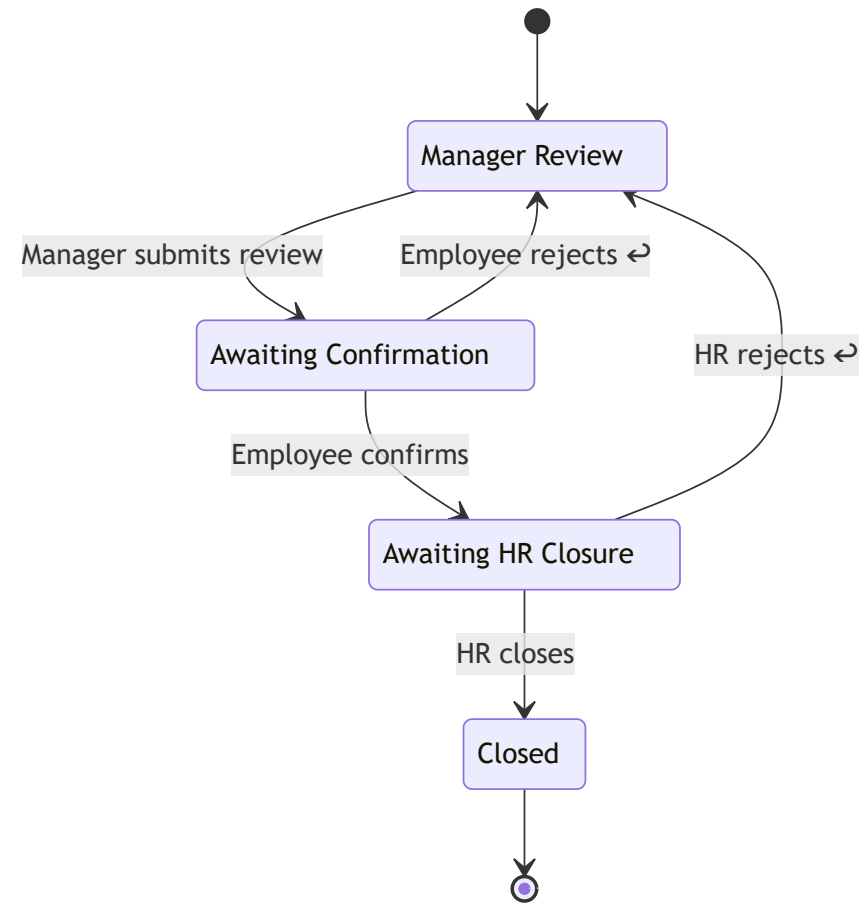


Evaluation Flow

Goal & Progress Phase

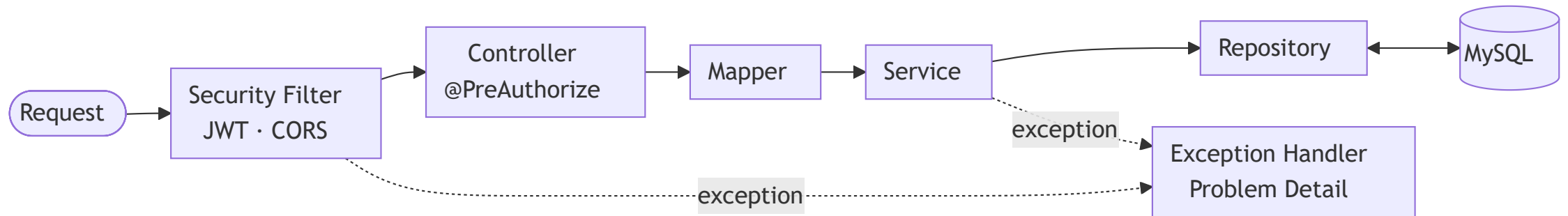


Review Phase



Application Architecture

Backend

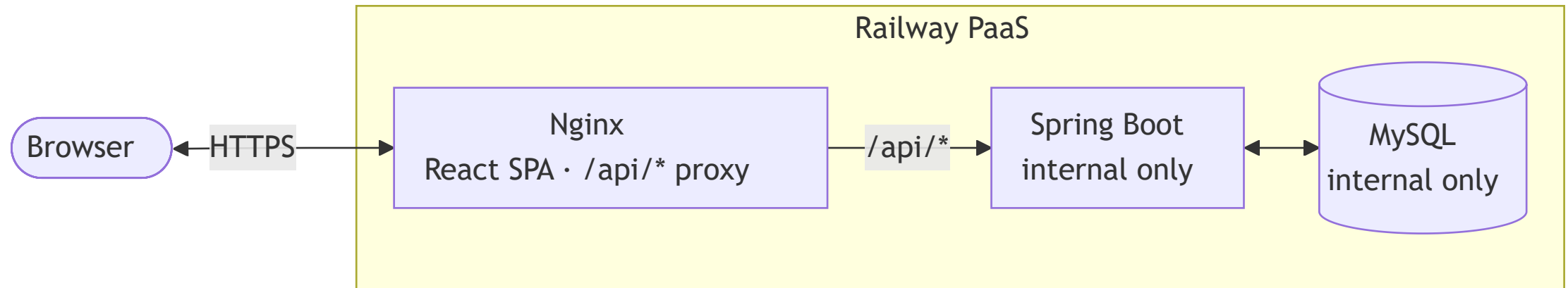


- Modules: `auth` · `user` · `evaluation` · `template` · `audit`

Frontend

- React SPA — Pages & reusable Components, with a TypeScript API client auto-generated by Orval from the OpenAPI spec.

System Architecture



- JWT stored in `HttpOnly; Secure; SameSite=Lax` cookie, preventing XSS & CSRF
- 2 replicas of backend & frontend respectively, behind Railway's load balancer

Unit Tests

Layer	Focus	Tool
Controller	Response format & exception shape	MockMvc
Service	Business logic & state transitions	Mockito
Repository	CRUD correctness	H2 + DataJpaTest
Frontend Section	Component behaviour per evaluation status	Vitest + RTL

- Coverage from SonarScanner: Backend 70.9%, Frontend 89.1%

Stress Test

Tool	k6
Scenario	Authenticate once, then repeatedly call <code>/me</code> and <code>/my-evaluations</code>
Load profile	50 → 250 → 500 → 1000 VUs, 30 min per stage
Metrics	Throughput (req/s), p95 latency per endpoint, error rate

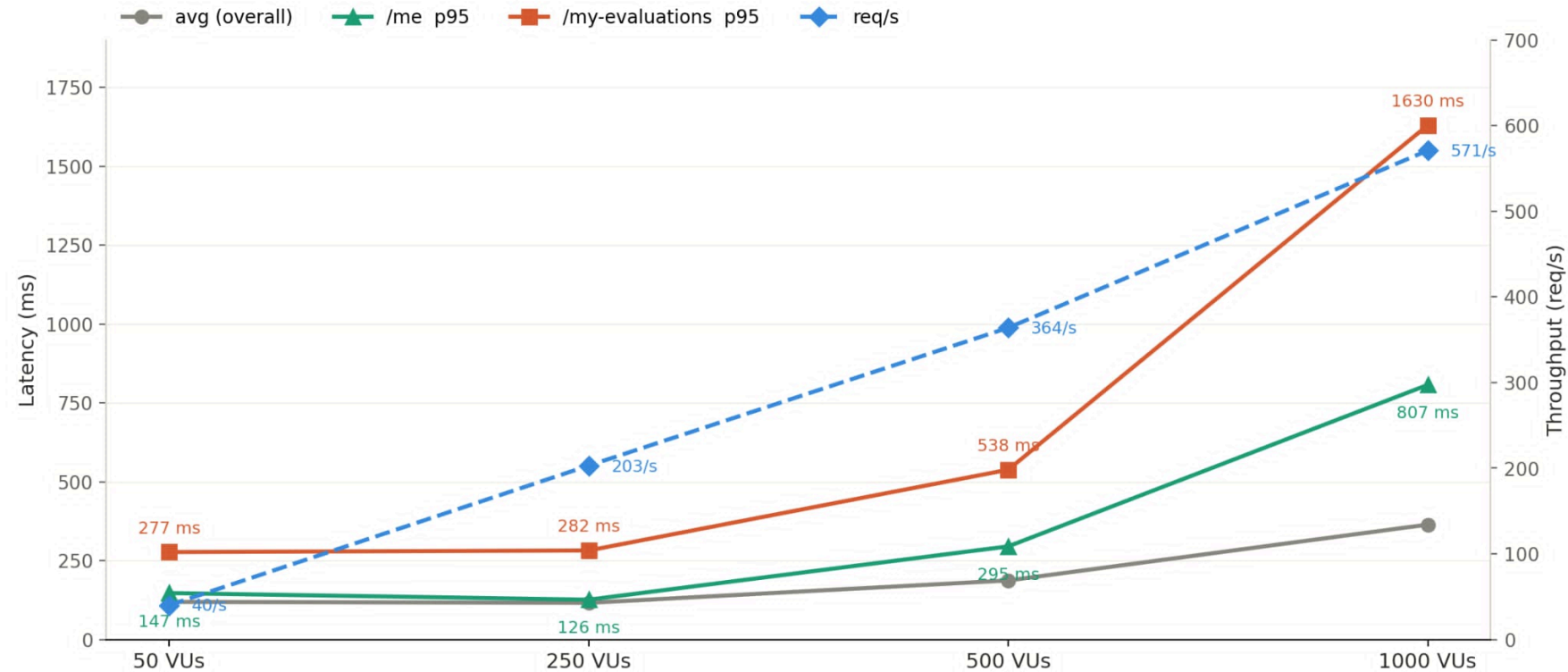
Stress Test Result

k6 Stress Test Results · 30-min sustained load

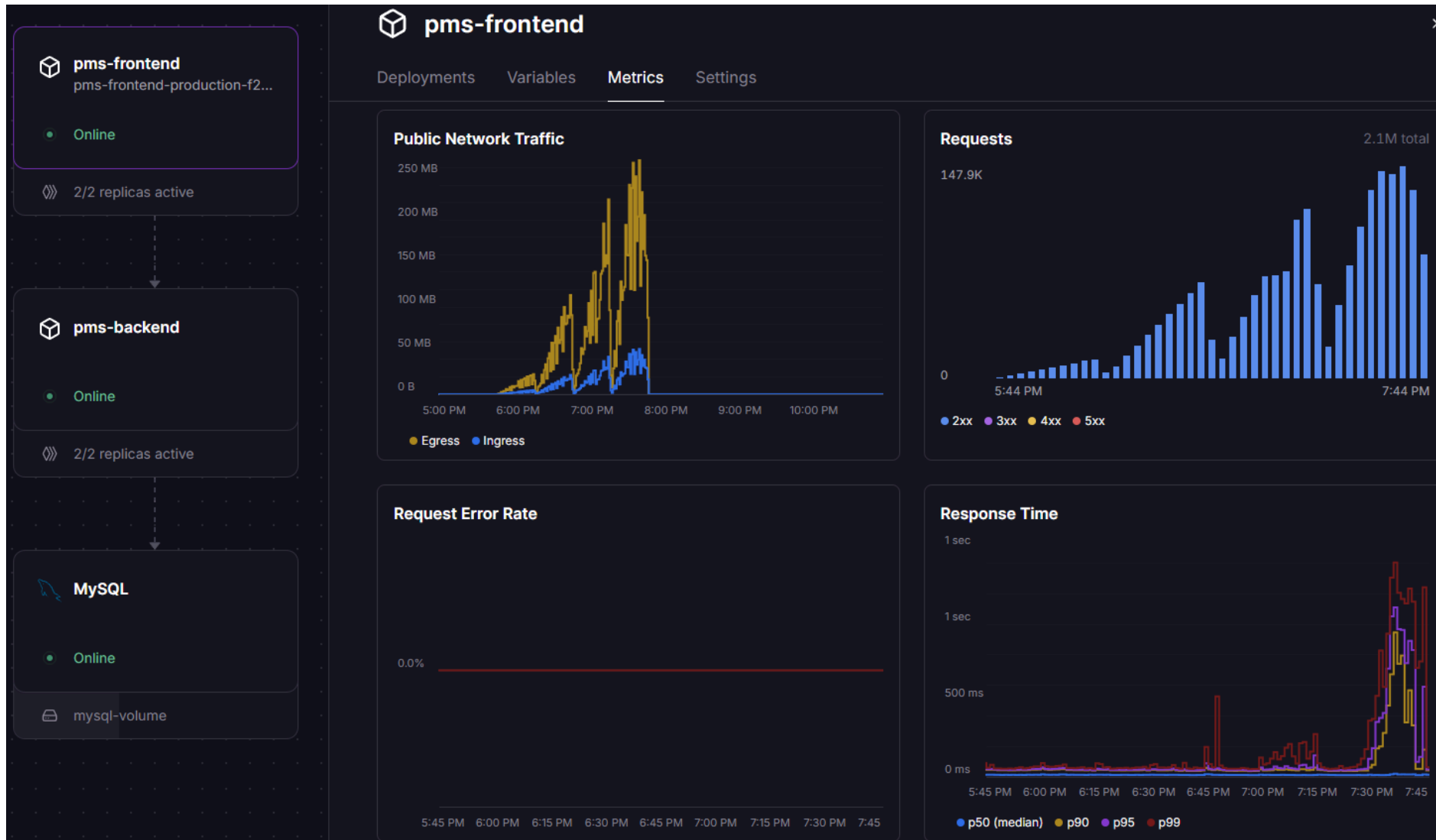
Peak RPS
571
@ 1000 VUs

Total Requests
2,137,091
across 4 runs

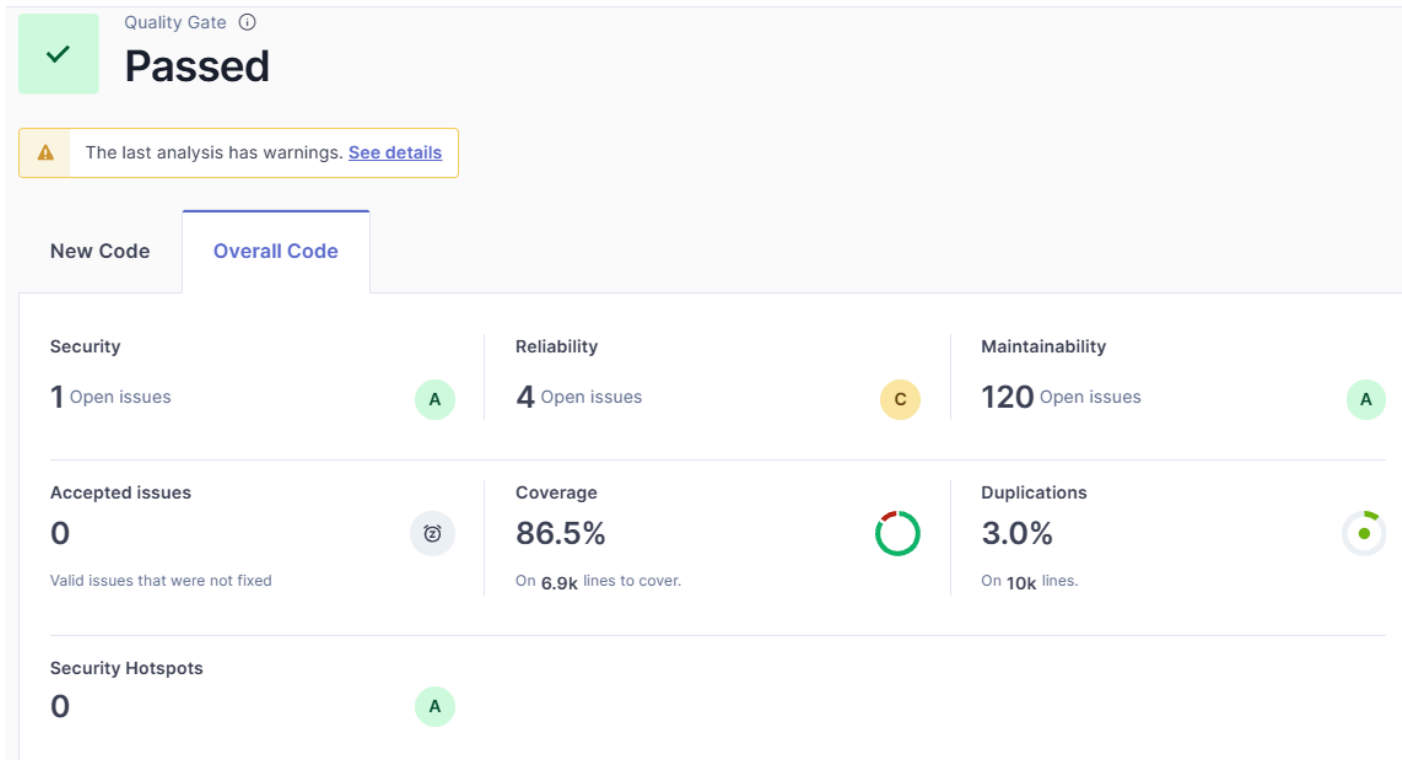
Max Error Rate
0.01%
@ 1000 VUs



Observability - Railway Built-in Dashboard



Code Quality



CI/CD — Tests + image build gate PR; Merge to `main` → GHCR → Railway redeploy

Demo

- Demo Link: <https://pms-frontend-production-f2a8.up.railway.app>
- Github Repo: <https://github.com/nora6633/ntu-cloudnative-pms>